Soham Sandeep Gurav

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Education

Pune Vidyarthi Griha's College of Engineering, Technology & Management 2025 – 2029 (Expected)
Bachelor of Technology in Artificial Intelligence and Data Science — *Pune, Maharashtra*

Maharashtra State Board of Secondary and Higher Secondary Education 2023 – 2025
HSC (Grade 12) – 77.83% — *Maharashtra*

Bharatiya Vidya Bhavan's Paranjape Vidya Mandir 2011 – 2023
SSC (Grade 10), Maharashtra State Board – 94.00% — *Kothrud, Pune*

Competitive Exams

MHT-CET 2025 & JEE Main 2025 2025
MHT-CET – 97.36 percentile — *JEE Main – 83.6 percentile*

IISER Aptitude Test (IAT) & NEST 2025
IAT Rank ∼60K — *Also appeared for NEST*

Experience

Fit Ez Sept 2026 – Present
Web Development & SEO Intern — *Remote*

- Building the website of a 1-on-1 diet coaching startup end to end, with the front end in HTML, CSS and JavaScript.
- Deployed the site on Cloudflare, connected the startup's domain, and set up redirects from old domains so existing links and traffic carried over.
- Working on search engine optimization so the site can be found by people searching for online coaching.

aIDEAS PVG (AI & DS Department Club) Aug 2026 – Present
Sponsorship & Public Relations Member — *Pune, Maharashtra*

- Write cold outreach emails, approach companies, and pitch the club's event to potential sponsors.
- Chose this team over a technical one to gain hands-on exposure to event finances, pitch framing for different audiences, and lead generation and follow-up.

IIT Madras CoERS Road Safety Hackathon May 2026 – July 2026
Team Lead, Copilot Crew — *Remote*

- Ideated and independently developed DriveLegal v1, owning its core architecture, technology choices and initial implementation.
- Led the submission (project code, YouTube demo video, presentation), and the team was selected for the final rounds.
- After selection, coordinated the team's work across UI development, leaderboard, JWT authentication and traffic-rule data collection.

GirlScript Summer of Code 2026 May 2026 – June 2026
Open Source Contributor — *Remote*

- Merged 6 PRs across 3 projects; held a global rank of 287 in the first weeks and finished at rank 2289 among 47,926 participants (top 5%).
- **Algo:** built persistent quiz and leaderboard data with Node.js, Express, SQLite and Sequelize; optimized images with Pillow/WebP and added dark mode support.
- **FinanceFlow:** built a topic-based learning architecture in Next.js using dynamic routes, centralized roadmap data and reusable components.
- **NPMChat:** improved type safety in a React/TypeScript chat app and standardized profile image handling across API payloads, controllers and database models.

DAUSS (Defense Aerial Unmanned Surveillance Sentinel) April 2026 – Present
Embedded Systems & AI Trainee — *Pune, Maharashtra*

- Member of the embedded systems team of the college club that designs and builds fixed-wing UAVs.
- Learning the hardware side of flight: working hands-on with UAV sensors and studying the CrossFlight microcontroller that ties them together.

YouTube April 2026 – Present
Content Creator *Independent*

- Run a channel that explains my projects in a narrative style: concepts, how they work, what I built, and where things broke, including failed attempts and debugging.
- Grown to 100+ subscribers; most watched video has crossed 300 views.

Projects

DriveLegal | *Python, FastAPI, Streamlit, Qdrant, FAISS, Ollama (Mistral), SQLite* **May 2026 – July 2026**

- Offline-capable, location-aware traffic-law assistant built with a Retrieval-Augmented Generation (RAG) pipeline to avoid hallucinated answers.
- Extracted official traffic-rule PDFs, chunked and embedded them into Qdrant, and passed the closest chunks to a locally run LLM as context.
- Used exact JSON data instead of vector search for the challan calculator, since fine amounts must be exact.
- Learned to swap to a local model (Mistral via Ollama) when API credits ran out, and to match storage choices to the job.

Stardust: TIR Image Colorization | *SRCNN, Pix2Pix GAN, Physics-Informed Loss* **2026**

- Round 1 submission to ISRO Bharatiya Antariksh Hackathon 2026 (PS10) on improving and colorizing Landsat thermal imagery; architecture and planning stage only, no working prototype.
- Led planning of a two-stage pipeline (SRCNN super-resolution, then a Pix2Pix GAN conditioned on vegetation and land-use data) and coordinated a four-person interdisciplinary team.

Doppler Speed Detector | *Arduino UNO, RCWL-0516, C++* **2026**

- Built a radar-inspired speed detector: the RCWL-0516 microwave sensor picks up the Doppler shift, the Arduino estimates its frequency, applies the Doppler formula, and shows speed in km/h on a 16×2 LCD.
- First hardware project; total cost under INR 900 and about two days of build time.

LockChain | *Solidity, Remix IDE, React, Web3, MetaMask, Firebase, Vercel* **2026**

- Built for the 24 Hours of Ignition Hackathon: a Solidity escrow smart contract (Sepolia testnet) that holds payment until agreed conditions are met, coordinating buyer, seller and release.
- React and Web3 front end connects wallets via MetaMask and listens for contract events; Firebase stores off-chain application state.

Validating the Venturi Ground Effect | *TinkerCAD, SimScale (CFD)* **2026**

- Modeled a ground-effect tunnel in TinkerCAD, built a cardboard prototype with a teammate, and ran the CFD simulation in SimScale (inlet/outlet boundary conditions, meshing).
- Validated the pressure drop physically (tissue paper sucked into the tunnel throat by a fan) and in simulation streamlines.

Physique Log | *HTML5, CSS3, JavaScript, Capacitor, Node.js* **2026**

- Solo-built, open-source standalone mobile app for fully customizable workouts (with personal diagrams) and daily progress tracking.
- Vanilla JS front end with browser localStorage, HTML5 Canvas charts, and Capacitor packaging for Android.

Technical Skills

Languages: Python, C++, Java, Arduino
Developer Tools: VS Code, Eclipse IDE, Jupyter Notebooks/Google Colab, GitHub, Git, Vercel, Firebase, Netlify, Supabase, Cloudflare
Technologies/Frameworks: Python Scrapy (web scraping), FastAPI, Docker, Qdrant, FAISS, Arduino, AI IDE Workflows, Python Poetry
Others: MathJax, LaTeX, Local LLMs, RAG